

RAILWAY NETWORK SIMULATION SYSTEM USER MANUAL

Integrative Project – Second Semester
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Glossary

Building	Edifício	Upgrade built on a station to improve its performance. Associated with a specific building type.
Building Type	Tipo de Edifício	Defines the effect and cost of a building. Examples include hotels, silos, cafés, etc. Can be affected by time restrictions.
Carriage	Carruagem	Part of the train responsible for carrying cargo. Varies depending on the type of cargo it transports and are free of costs of purchase.
Cargo	Cargo	List of things carried by the trains, from resources to final products.
Cargo Mode	Modo de Carga	Defines the loading condition that dictates when a train departs from a Point-of-Route based on the amount of cargo loaded: full, half or available.
Cargo Status	Estado da Carga	Indicates whether a cargo is in supply or demand at a given point in the simulation.
City	Cidade	Landmark formed by a gathering of blocks that serve as houses. In this context, works as a consumer and/or a generator.
City Block	Bloco da Cidade	Element that forms a city. It consumes and generates cargo, such as passengers or mail.
Demand	Procura	Business received by industries or cities depending on the radius of the associated station.
"Depot"	"Armazém"	Stations with fifty thousand capacity and a three by three radius.

Editor	Editor	User who is able to create maps and scenario from an included bundle or an exclusively sold one.
Final product	Produto final	Range of products, such as food, textiles or cars, made in industries that ended up being consumed by cities.
Historical Restriction	Restrição Histórica	Restrictions related to the historical events which may influence the simulation.
Industry	Indústria	Landmarks where products are made or transformed, available in three different types.
Industry Changer	Modificador de Indústria	Alters the behavior or availability of industries, typically due to historical restrictions.
Locomotive	Locomotiva	Part of the train that provides traction to the train. It can run by steam, diesel or electricity and is distinguished by its model.
Locomotive Model	Modelo de Locomotiva	Blueprint for locomotives, defining their characteristics like power, cost, fuel consumption, and year of operation.
Locomotive Type	Tipo de Locomotiva	Categorization of locomotives (e.g., steam, diesel, electric). Subject to technological restrictions.
Mail	Correio	One of the different things consumed and generated by cities.
Map	Mapa	Spatial representation containing cities and industries. Created by an Editor and used in scenarios.
Mixed industry	Indústria Mista	Ports where cargo is either imported or exported.

Passenger	Passageiro	One of the various things consumed and generated by cities.
Player	Jogador	User who is only able to play the game, whether single or multiplayer.
Primary sector industry	Indústria de setor primário	Industries that generate either mineral or farming resources.
Point of Route	Ponto da Rota	A segment of a route where the train stops at a specific station and attempts to pick up a predefined list of cargoes. Each point also specifies a cargo mode
Ports Behaviour	Comportamento dos Portos	Defines which resources can be imported/exported and which industry types are affected.
Position	Posição	Coordinates that define the location of a station, city, or industry on the map.
Product	Produto	Come from Transforming industries and can be used again to make final products in a different industry.
Railway	Ferrovia	The metal tracks on which trains run.
Railway Line	Linha ferroviária	Path that connects to stations, with different lengths, sizes and electrification.
Railway Type	Tipo de Ferrovia	Defines the cost and whether the railway is electrified. Each railway segment is associated with one type.
Resource	Recurso	Generated in Primary sector industries, end up being consumed or processed in other industries.
Resource Type	Tipo de Recurso	Classification of a cargo resource, indicating its value and origin (e.g., minerals, farm products).

Route	Rota	A list of Points of Route that a train follows. It specifies the sequence of stations the train will visit and the cargo operations to be performed at each stop.
Scenario	Cenário	A setup defined by an Editor which includes a map, stations, rules, restrictions, and other simulation parameters.
Scenario Editor	Editor de cenário	Tool that enables the user to apply restrictions or conditions for using a map.
Simulator	Simulador	The core system used by the player to control and monitor the simulation, including routes, trains, and performance metrics.
Station	Estação	Landmarks that serve cities and industries connected by lines, where trains collect and deliver their assigned cargo. Can appear in three different types.
"Station"	"Estação"	Stations with one hundred thousand capacity and a four by four radius.
Station Type	Tipo de Estação	Defines radius and cost of a station. Each station must be associated with one type (e.g., Depot, Station, Terminal).
Supply	Abastecimento	Business sent by industries or cities depending on the radius of the associated station.
System User	Utilizador do sistema	Type of user operating the system, either Editor or Player.
Technological Restriction	Restrição Tecnológica	Restrictions related to the machinery (industries and locomotives) available.

"Terminal"	"Terminal"	Stations with two hundred thousand capacity and a five by five radius.
Time Restriction	Restrição de Tempo	Restrictions related to the time period where the simulation will take place.
Train	Comboio	Composed of a locomotive and carriages. Assigned to a route and used to transport cargo between stations.
Transforming industry	Indústria Transformadora	Industries, such as bakeries, textiles, automobiles or steel mills, that transform the resources into new products.
Train	Comboio	Composed of a locomotive and carriages. Assigned to a route and used to transport cargo between stations.



Introduction

Welcome to the User Manual for the Railway Network Simulation System, created within the scope of the Integrative Project. This guide aims to help users understand and operate the system's key features and functionalities, enabling them to navigate and make full use of the software developed during the semester.

Purpose and Scope

The objective of this manual is to provide clear, accessible instructions for using the railway network simulation platform. It covers both the map and scenario editor and the simulation environment, offering users the tools to design, manage, and analyse virtual railway systems. Whether the user is building a fictional city network or testing logistical strategies, this guide ensures a smooth and informed experience with the platform.

The product described in this manual is an educational software solution inspired by classic simulation games like Railroad Tycoon. It aims to simulate the operation of railway networks in a simplified yet insightful way. Users can create maps with cities and industries, define historical and technological scenarios, and manage stations, trains, and routes. Additionally, it includes tools for statistical analysis and topological validation, allowing users to assess the performance and connectivity of the network.

Target Audience

This manual is intended for two types of users:

- **Editors:** Users who create and configure maps and scenarios. This role involves defining the geographical and temporal context of the simulation.
- **Players:** Users who engage with the simulation, making strategic decisions such as building stations, buying locomotives, and setting train routes. In future iterations, the platform may support multiplayer experiences.

Whether you're a student building the project, a professor reviewing its progress, or a user testing the final product, this manual aims to offer a comprehensive and user-friendly overview of the platform. Let's begin your journey into the dynamic world of railway management.

System Overview

Purpose and Use of the Product

The application is a software platform that integrates a **map and scenario editor** with a **railway operation simulator**. It is designed for educational, analytical, and entertainment purposes. Users can design geographical maps with cities and industries, define simulation scenarios with constraints, and simulate the operation of a railway network by building infrastructure, managing trains, and transporting cargo.

The application serves three main types of users:

- **Editors**, who creates and configure maps and scenarios.
- **Players**, who operates the railway simulation.
- **Product Owners**, who evaluate the system's behaviour, performance, and algorithms.

Structure of the Product

The system is divided into two primary components:

1. Map and Scenario Editor

- Allows users to create grid-based maps.
- Supports placement of cities and industries.
- Defines timeframes, technological restrictions, and historical events relevant to the simulation.

2. Simulator

- Allows players to build railway infrastructure (stations and lines).
- Simulates train operation and cargo transport between stations.
- Handles the generation and delivery of cargo, considering demand, station upgrades, and available resources.
- Provides options to play, pause, and analyse the simulation dynamically.

These components rely on a **shared domain model** that includes maps, cities, industries, stations, locomotives, and cargo types. The simulator also integrates with modules for network analysis and statistical evaluation.

The system is implemented in **Java**, with the graphical interface built using **JavaFX 11**, and statistical analysis carried out using **Python (Jupyter Notebooks)**.

Reference to Main Features

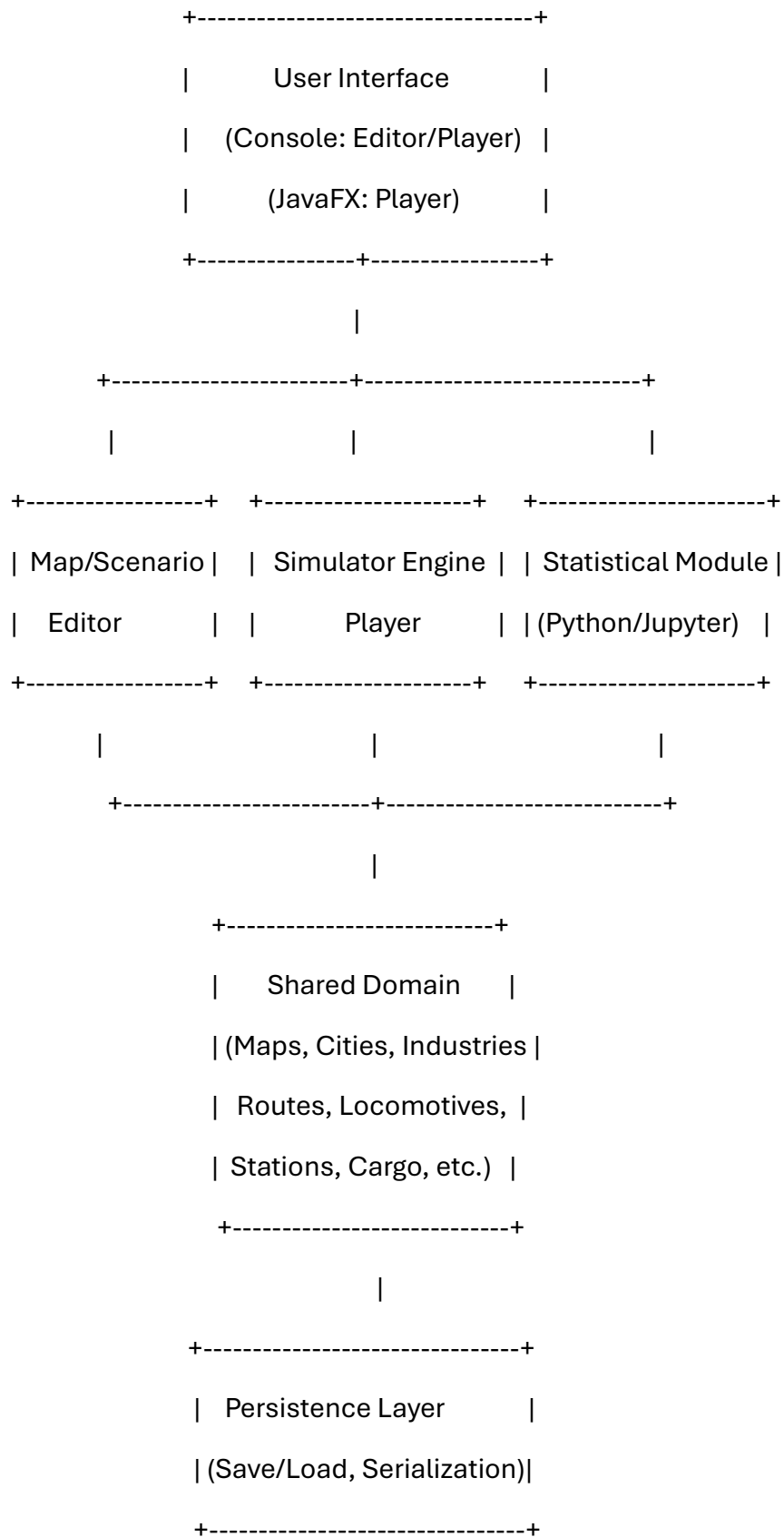
The application includes the following key features:

- Editable maps with cities and industries.
- Scenario configuration with technological and historical parameters.
- Construction and upgrade of three station types (Depot, Station, Terminal).
- Locomotive acquisition and route planning with cargo loading options.
- Simulation of train operations, cargo delivery, and economic outcomes.
- Visual representation of the railway network and connectivity.
- Shortest route calculation and maintenance path algorithms.
- Statistical tools for analysing station performance and cargo patterns.
- Save and load functionalities for maps, scenarios, and game states.
- Secure user authentication with strong password requirements.

Highlight of the Main Modules

Module	Description
-----	-----
Map Editor	Enables creation of grid maps with cities and industries.
Scenario Editor	Defines simulation constraints and scenario parameters.
Station Manager	Builds and upgrades stations with economic impact.
Route Planner	Assigns trains to custom-defined travel routes.
Simulator Engine	Simulates train movement, cargo transport, and economic transactions.
Network Analyzer	Validates train connectivity and finds routes for maintenance or optimization.
Statistics Engine	Conducts analysis using Jupyter Notebooks and visualizations.
Persistence Layer	Supports saving and loading all game and simulation data.

Diagram: Product Structure Overview



System Requirements

This section outlines the technical requirements necessary for installing, running, and maintaining the application. It includes information on compatible operating systems, hardware resources, dependencies, and installation steps.

Compatible Operating Systems

The application is developed in Java and is compatible with the following operating systems:

- **Windows 10 or higher** (64-bit, with latest updates recommended)
- **macOS 11.0 (Big Sur)** or later
- **Linux distributions** with Java 17+ support (e.g., Ubuntu 20.04 LTS or later)

Required Software and Plugins

To run the application and its statistical analysis modules, the following software must be installed:

Required Runtime Environments

- **Java Development Kit (JDK):** Version 17 or higher
- **JavaFX 11+ SDK:** Required for the graphical user interface
- **Python 3.8+:** Required for statistical analysis notebooks
- **Jupyter Notebook:** Required for executing Python-based analysis
- **Graphviz or GraphStream:** Optional but recommended for graph visualizations in connectivity and maintenance analysis

Optional Tools (for development)

- **IntelliJ IDEA or Eclipse IDE**
- **JUnit 5** (for running unit tests)
- **JaCoCo Plugin** (for code coverage reports)

Hardware Requirements

Component	Minimum Requirement	Recommended
CPU	Dual-core 2.0 GHz	Quad-core 2.5 GHz or better
RAM	2 GB	4 GB or more
Disk Space	500 MB free space	1 GB free space
Screen Resolution	1280x720 (HD)	1920x1080 (Full HD)
GPU	Integrated (JavaFX-compatible)	Dedicated GPU (for smooth GUI)

Note: Additional space is needed for simulation data, logs, saved files, and optional assets.

Installation Procedures

1. **Download** the application package from the official repository or release page.
2. **Install Java 17+** and **JavaFX 11+** if not already present.
3. **Set Environment Variables:** Ensure JAVA_HOME and JavaFX module paths are configured (if launching from terminal/CLI).
4. **Run the Installer** (if provided) or launch the .jar file using a command
5. **For Python-based analysis:**
 - Install Python 3.8+
 - Use “pip install jupyter pandas matplotlib seaborn” to install dependencies
 - Launch Jupyter Notebook and open the “.ipynb” file provided with the statistical module

Other Technical Specifications

- The system uses **object serialization** for persisting data across sessions.
- Simulation data and save files are stored locally using a structured directory.
- All configuration and documentation files should be saved using **UTF-8 encoding**.
- All diagrams and figures generated during development or simulation analysis should be saved in **SVG format**.
- The system enforces a **user login mechanism** with password validation (7-character alphanumeric, including at least three uppercase letters and two digits).

Features

This section provides a detailed overview of the application's main functionalities, organized according to the user stories defined in the project. Each feature includes the role responsible for using it, a clear title, and step-by-step instructions to ensure ease of use. Where applicable, visual aids such as screenshots or diagrams are recommended to support user understanding. These features collectively support the creation, configuration, simulation, and analysis of a railway network scenario.



User Login

User Role: Editor or Player

Steps:

1. Choose the role (Editor or Player).
2. Type a username (must be valid characters only).
3. Type a password (must have 7 digits, including 3 uppercase letters and 2 numbers).
4. Type a budget (Player only).
5. Choose between graphical interface or console (Player only).

```
=== Login Menu ===  
1. Login as Editor  
2. Login as Player  
0. Exit
```

Figure 1: Login Menu

```
=== Editor Menu ===  
1. Create Map  
2. Create Industry  
3. Add City  
4. Create Scenario  
5. Save Map  
6. Load Map  
7. Save Scenario  
8. Load Scenario  
0. Exit
```

Figure 2: Editor Menu

```
Player Menu  
  
Build Station  
  
Upgrade Station  
  
View Station Details  
  
Buy Locomotive  
  
Assign Route  
  
Simulator  
  
Logout
```

Figure 3: Player Menu

Create a Map

User Role: Editor

Steps:

1. Login in the Editor Menu.
 2. Choose "Create New Map".
 3. Enter the map name (must be a valid file name).
 4. Define a size (in positive integers).
 5. Set the map scale (e.g., 1 cell = 10 km).
 6. Save the map.
-

Create an Industry

User Role: Editor

Steps:

1. Login in the Editor Menu.
2. Choose "Create Industry".
3. Select a map.
4. Type a desired XY position (respecting the map size).
5. Choose from the list of available industry types.
6. Save the industry in the map.

Add a City

User Role: Editor

Steps:

1. Login in the Editor Menu.
 2. Choose "Add City".
 3. Select a map.
 4. Enter the name (letters only).
 5. Type a desired XY position (respecting the map size).
 6. Set number and position of house blocks (manual or automatic).
 7. Save the city in the map.
-

Create a Scenario

User Role: Editor

Steps:

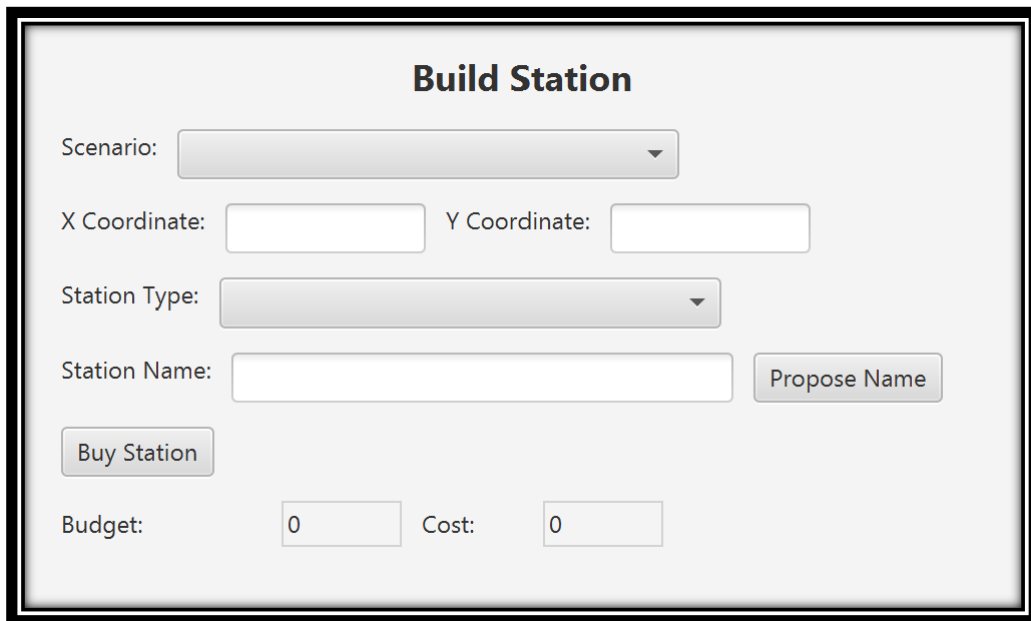
1. Login in the Editor Menu.
2. Choose "Create Scenario".
3. Select a map.
4. Define simulation start year.
5. Configure available locomotives.
6. Set port behaviors (resources imported/exported).
7. Define industry behaviour (generation rate).
8. Save the scenario.

Build a Station

User Role: Player

Steps:

1. Login in the Player Menu.
2. Choose "Build Station".
3. Select a scenario.
4. Type a desired XY position (respecting the map size).
5. Choose a station type (Depot, Station, Terminal).
6. Type a center (if Station is chosen).
7. Accept system-proposed name or type one.
8. Confirm build.
9. Save the station in the scenario.



The screenshot shows a web-based interface titled "Build Station". It contains the following elements:

- A "Scenario:" dropdown menu.
- Input fields for "X Coordinate:" and "Y Coordinate:".
- A "Station Type:" dropdown menu.
- A "Station Name:" input field and a "Propose Name" button.
- A "Buy Station" button.
- Input fields for "Budget:" (containing the value 0) and "Cost:" (containing the value 0).

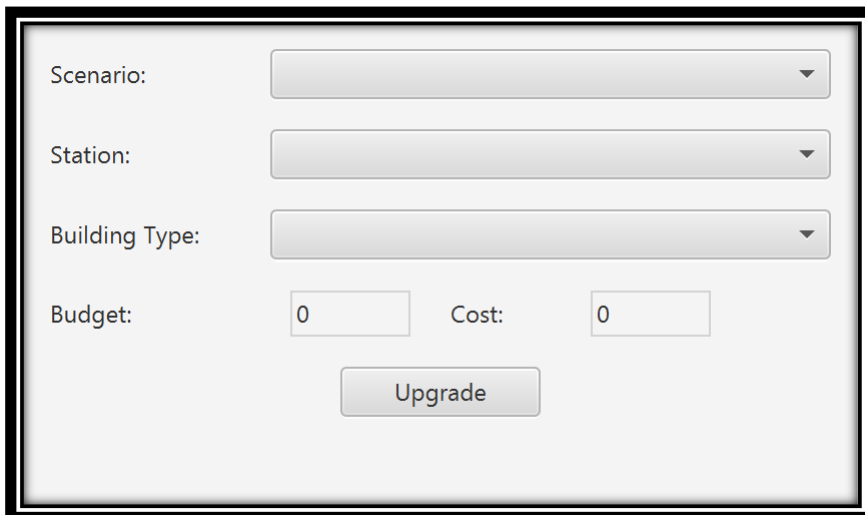
Figure 4: Build Station Menu

Upgrade a Station

User Role: Player

Steps:

1. Login in the Player Menu.
2. Choose "Build Station".
3. Select a scenario.
4. Select the station.
5. Select the building type from the available ones.
6. Confirm upgrade.
7. Save the building in the station



The screenshot shows a user interface for upgrading a station. It features four dropdown menus labeled 'Scenario:', 'Station:', and 'Building Type:'. Below these are two input fields for 'Budget:' and 'Cost:', both containing the value '0'. An 'Upgrade' button is positioned at the bottom center of the form.

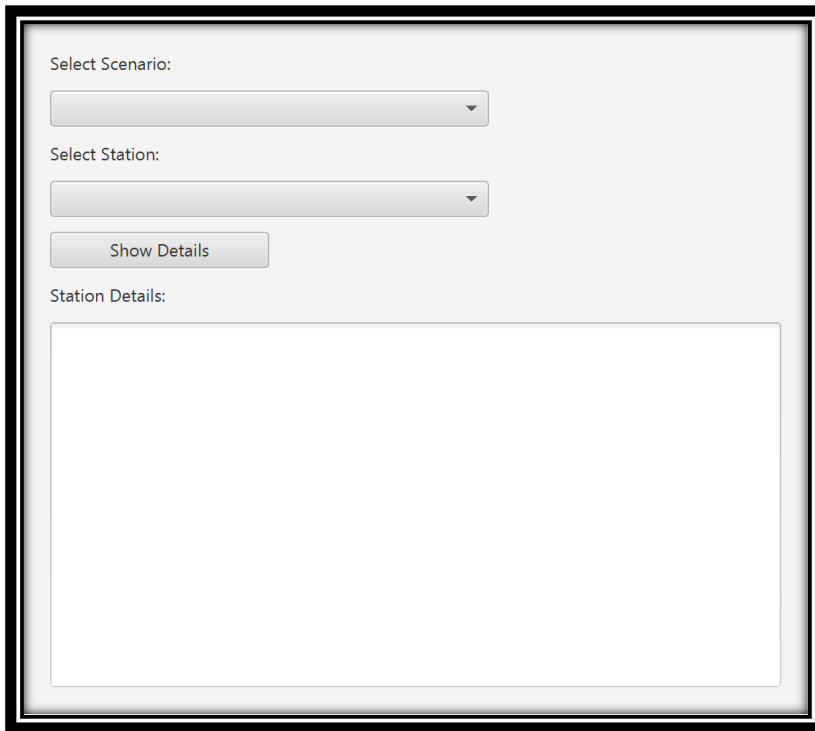
Figure 5: Upgrade Station Menu

View Station Details

User Role: Player

Steps:

1. Login in the Player Menu.
2. Choose "Build Station".
3. Select a scenario.
4. Select the station.
5. View existing buildings and cargo demand/supply.



The screenshot shows a web interface for viewing station details. It features two dropdown menus: 'Select Scenario:' and 'Select Station:'. Below these is a 'Show Details' button. At the bottom, there is a large empty rectangular box labeled 'Station Details:'.

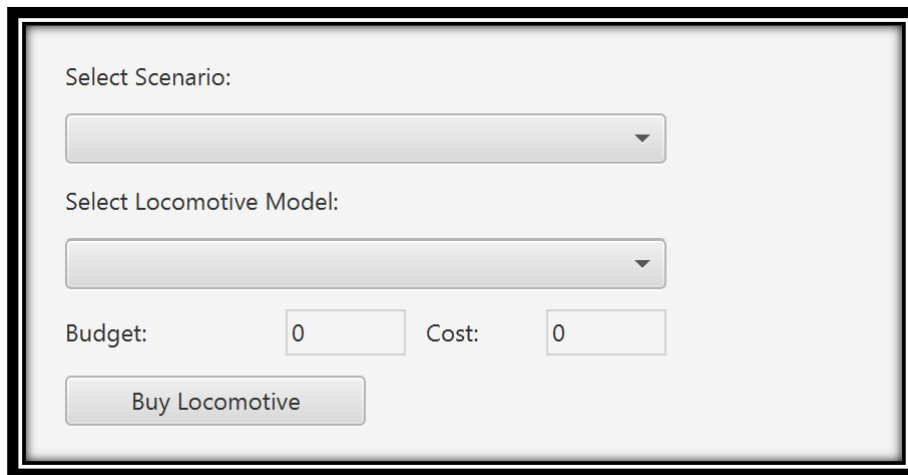
Figure 6: View Station Details Menu

Buy a Locomotive

User Role: Player

Steps:

1. Login in the Player Menu.
2. Choose "Build Station".
3. Select a scenario.
4. Select a model from the available ones.
5. Confirm purchase.
6. Save the locomotive in the scenario.



The screenshot shows a user interface for purchasing a locomotive. It features two dropdown menus for selecting a scenario and a locomotive model. Below these are input fields for 'Budget' and 'Cost', both currently set to '0'. A 'Buy Locomotive' button is positioned at the bottom of the form.

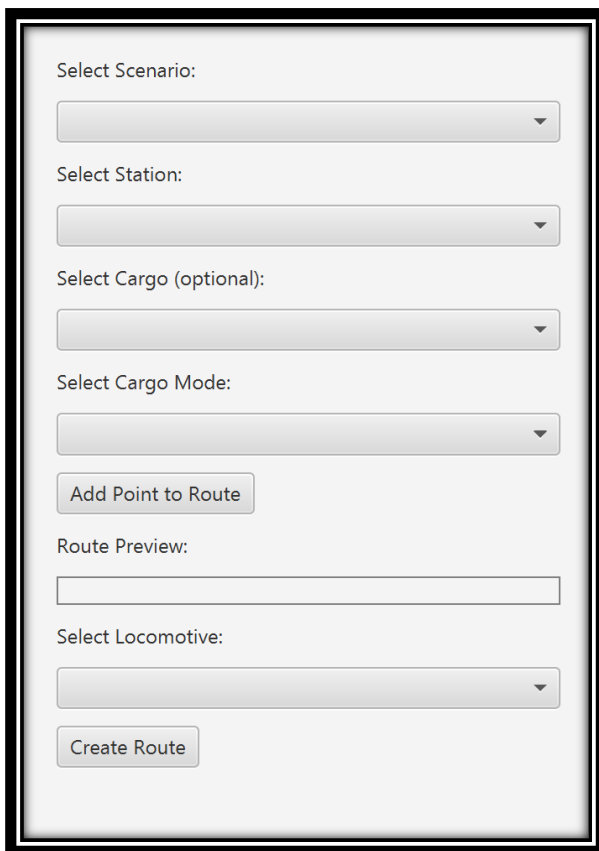
Figure 7: Buy Locomotive Menu

Assign a Route

User Role: Player

Steps:

1. Login in the Player Menu.
2. Choose "Assign a Route".
3. Select a scenario.
4. Type the number of points of route.
5. Select a station.
6. Select cargo from the available.
7. Select a mode.
8. Select a locomotive
9. Save the route in the scenario.



The screenshot shows a web-based interface for assigning a route. It contains several dropdown menus and two buttons. The dropdowns are labeled: 'Select Scenario:', 'Select Station:', 'Select Cargo (optional):', 'Select Cargo Mode:', and 'Select Locomotive:'. The buttons are labeled 'Add Point to Route' and 'Create Route'. There is also a 'Route Preview:' label above a text input field.

Figure 8: Assign Route Menu

Simulator

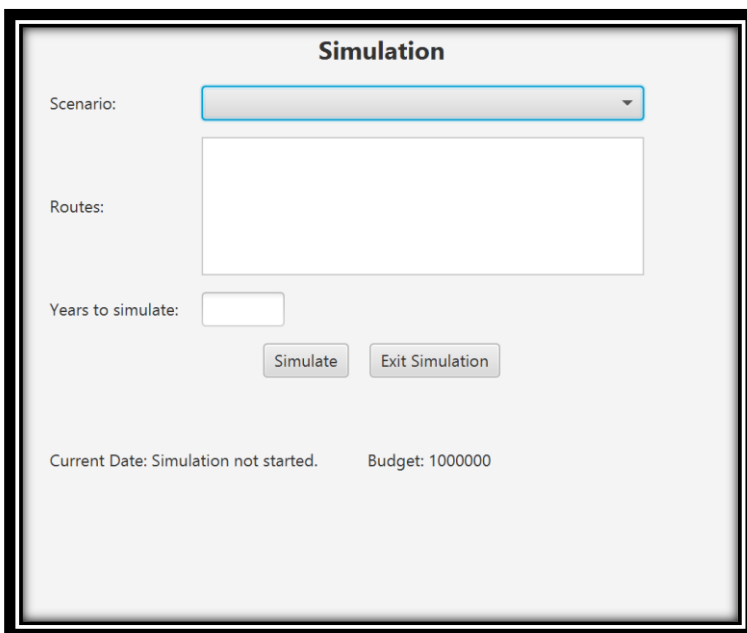
User Role: Player

Steps:

1. Login in the Player Menu.
2. Choose "Simulator".
3. Choose to either create a simulation or continue one
4. Select a scenario.
5. Select the routes wanted in the simulation.
6. Select how many years to simulate.
7. Simulate until request to stop.

```
=== Simulator Menu ===
1. Create New Simulation
2. Check Simulation Status
3. Continue Existing Simulation
4. View Financial Report
0. Back to Main Menu
```

Figure 9: Simulator Menu



The screenshot shows a web-based interface titled "Simulation". It contains the following elements:

- A "Scenario:" label followed by a dropdown menu.
- A "Routes:" label followed by a large, empty text input area.
- A "Years to simulate:" label followed by a small text input field.
- Two buttons: "Simulate" and "Exit Simulation".
- At the bottom, two status indicators: "Current Date: Simulation not started." and "Budget: 1000000".

Figure 10: Simulation Menu

Save Map

User Role: Editor

Steps:

1. Login in the Editor Menu.
 2. Choose “Save Map”.
 3. Select a Map.
 4. Write the serialized file name.
-

Load Map

User Role: Editor

Steps:

1. Login in the Editor Menu.
 2. Choose “Load Map”.
 3. Write the name of the path to the file.
-

Save Scenario

User Role: Editor

Steps:

1. Login in the Editor Menu.
 2. Choose “Save Scenario”.
 3. Select a Scenario.
 4. Write the serialized file name.
-

Load Scenario

User Role: Editor

Steps:

1. Login in the Editor Menu.
 2. Choose “Load Scenario”.
 3. Write the name of the path to the file.
-

Save Simulation

User Role: Player

Steps:

1. Login in the Player Menu.
 2. Choose “Save a Simulator”.
 3. Select a Simulator.
 4. Write the serialized file name.~
-

Load Simulation

User Role: Player

Steps:

1. Login in the Player Menu.
 2. Choose “Load a Simulator”.
 3. Write the name of the path to the file.
-

View Financial Report

User Role: Player

Steps:

1. Login in the Player Menu.
 2. Choose “Simulator Menu”.
 3. Choose “View Financial Report”.
-

Troubleshooting

This section provides a list of possible issues that users may encounter while using the Railway Network Simulation System. For each issue, the symptom, likely cause, and suggested solution are provided to help resolve the problem efficiently.

Issue 1 – Application does not open

- **Symptom:** Double-clicking the application icon has no effect.
- **Possible Cause:** Java runtime environment is not installed or not correctly configured.
- **Solution:**
 1. Verify that Java 17 or higher is installed.
 2. Check that the JAVA_HOME environment variable is correctly set.
 3. Reboot the system and try again.
 4. If the issue persists, try launching the application via the terminal to view error messages.

Issue 2 – Missing JavaFX modules

- **Symptom:** Error message: java.lang.module.FindException: Module javafx.controls not found.
- **Possible Cause:** JavaFX SDK is not configured in the classpath/module path.
- **Solution:**
 1. Ensure JavaFX 11+ SDK is downloaded.
 2. Run the application with the correct --module-path and --add-modules flags.
 3. Follow the installation instructions in Section 3.4 of this manual.

Issue 3 – Cannot save/load files

- **Symptom:** Saving or loading maps, scenarios, or simulations fails.
- **Possible Cause:** Application lacks permission to access the file system.
- **Solution:**
 1. Make sure the application is running with sufficient permissions.
 2. Check the folder paths are valid and writable.
 3. Avoid using special characters in file names.

Issue 4 – Python Notebook does not run

- **Symptom:** Code cells in the Jupyter Notebook fail to execute.
- **Possible Cause:** Required Python packages are not installed.
- **Solution:**
 1. Open terminal or command prompt.
 2. Run: “pip install jupyter pandas matplotlib seabornReposition”.
 3. Restart Jupyter Notebook and try again.

Need More Help?

If you continue to experience difficulties, please contact your instructor or the technical support team.

- **Helpdesk Email:** support@thbigbassers.xyz
- **Available:** Monday–Friday, 9:00–17:00
- **Phone:** +351 123 456 789

Frequently Asked Questions

Q.1: What file format is used to save and load scenarios/maps and simulations?

A.1: Scenario/maps and simulations are stored in serialized .ser files, which include all required parameters.

Q.2: Can I edit a scenario after saving it?

A.2: No. To modify a scenario, you must reconfigure the elements and save it again as a new .ser file.

Q.3: How do I know which cargo affects revenue the most?

A.3: Use the statistical analysis tool (US31). It automatically calculates correlations and shows which cargo is most influential.

Q.4: Can I run multiple simulations at the same time?

A.4: No. Only one simulation instance is supported at a time per application window.

Q.5: Can I recover a simulation if I accidentally overwrite it?

A.5: No, unless you have backups. It's recommended to use different file names for each version of a simulation.